Yuhui Hong

Last updated: October 14, 2026

Contact

Foege Building, Room S340A
University of Washington
Seattle, WA 98195

yuhhong@uw.edu
josiehong.github.io

Research interests

I develop AI methods for molecular discovery through computational mass spectrometry. My research focuses on molecular representation learning, molecular identification beyond reference libraries, and reliable inference from experimental spectra.

Appointments

University of Washington Aug. 2025–present
Postdoctoral Scholar, Department of Genome Sciences
Advisor: William Stafford Noble

Education

Indiana University Bloomington Sep. 2020–Jul. 2025
Ph.D. in Computer Science
Advisor: Haixu Tang
Dissertation: "Machine Learning Methods for Small Molecule Analysis in LC-MS/MS"

Xidian University Sep. 2015–Jul. 2019
B.S. in Computer Science and Technology

Publications

First author

1. **Yuhui Hong**, Sujun Li, Yuzhen Ye, & Haixu Tang (2025). FIDDLE: a deep learning method for chemical formulas prediction from tandem mass spectra. *Nature Communications*, 16(1), 11102.
2. **Yuhui Hong**, Yuzhen Ye, & Haixu Tang (2025). Machine learning in small-molecule mass spectrometry. *Annual Review of Analytical Chemistry*, 18(1), 193–215.
3. **Yuhui Hong**, Christopher J. Welch, Patrick Piras, & Haixu Tang (2024). Enhanced structure-based prediction of chiral stationary phases for chromatographic enantioseparation from 3D molecular conformations. *Analytical Chemistry*, 96(6), 2351–2359.
4. **Yuhui Hong**, Sujun Li, Christopher J. Welch, Shane Tichy, Yuzhen Ye, & Haixu Tang (2023). 3DMolMS: prediction of tandem mass spectra from 3D molecular conformations. *Bioinformatics*, 39(6), btad354.

Contributing author

6. Ludwig Lautenbacher, Kevin L. Yang, Tobias Kockmann, Christian Panse, Wassim Gabriel, Dulguun Bold, Elias Kahl, Matthew Chambers, Brendan X. MacLean, Kai Li, Fengchao Yu, Brian C. Searle, Damien Beau Wilburn, Mohammad Reza Zare Shahneh, **Yuhui Hong**, Haixu Tang, Mingxun Wang, Ralf Gabriels, Robbin Bouwmeester, Robbe Devreese, Jesse Angelis, Eduard Sabidó, Tobias K. Schmidt, Alexey I. Nesvizhskii, & Mathias Wilhelm (2025). Koina: democratizing machine learning for proteomics research. *Nature Communications*, 16(1), 9933.
7. Mahsa Monshizadeh*, **Yuhui Hong***, & Yuzhen Ye (2025). Multitask knowledge-primed neural network for predicting missing metadata and host phenotype based on human microbiome. *Bioinformatics Advances*, 5(1), vbae203. (*Co-first authors)

8. Yifan Zhang, Zhaojie Hu, Xueqiang Wang, **Yuhui Hong**, Yuhong Nan, XiaoFeng Wang, Jiatao Cheng, & Luyi Xing (2024). Navigating the Privacy Compliance Maze: Understanding Risks with Privacy-Configurable Mobile SDKs. *33rd USENIX Security Symposium (USENIX Security 24)*, 6543–6560.
9. Yaochen Li, **Yuhui Hong**, Yonghong Song, Chao Zhu, Ying Zhang, & Ruihao Wang (2022). SiamPolar: Semi-supervised realtime video object segmentation with polar representation. *Neurocomputing*, 467, 491–503.
10. Yaochen Li, Chao Zhu, Yuehu Liu, **Yuhui Hong**, & Jianji Wang (2021). Geometric and semantic analysis of road image sequences for traffic scene construction. *Neurocomputing*, 465, 336–349.

Preprints

1. Xianghu Wang, Abubaker Patan, Haoqi Nina Zhao, Vincent Charron-Lamoureux, Daniel Petras, **Yuhui Hong**, Benjamin P. Bowen, Trent Northen, Pieter C. Dorrestein, & Mingxun Wang (2026). Agentic AI for Structural Elucidation and Discovery of Drug Metabolites from Mass Spectrometry Data. (Under review).
2. **Yuhui Hong**, & Haixu Tang (2025). A Task-Specific Transfer Learning Approach to Enhancing Small Molecule Retention Time Prediction with Limited Data. (Under review).
3. **Yuhui Hong***, Mahsa Monshizadeh*, & Yuzhen Ye (2025). Confounder Free Predictive Models for Microbiome-based Host Phenotype Prediction. (Under review; *Co-first authors).

Books and patent

1. Haixu Tang, **Yuhui Hong**, & Sujun Li (2025). Method of predicting MS/MS spectra and properties of chemical compounds. U.S. patent application publication US20250356958A1.
2. Qingyang Xiao, Kaiyuan Liu, **Yuhui Hong**, & Haixu Tang (2024). Neural Networks for Chemists. *American Chemical Society*, DOI:10.1021/acsinfocus.7e8012.

Honors and fellowships

- UW Data Science Fellowship, eScience Institute 2025
- Luddy Outstanding Research Award, Indiana University Bloomington 2025
- Meritorious Winner, Mathematical Contest in Modeling 2018
  Top 10% among 8,085 teams worldwide.

Teaching experience

Indiana University Bloomington

- **Course Designer**, DSCI-D590, AI on Ramp Fall 2025
  Designed course content and assignments on convolutional neural networks, graph neural networks, transformers, and reinforcement learning. Materials continue to be used in subsequent semesters.
- **Instructor**, DSCI-D590, Topics in Data Science Spring 2025
  Taught course content and guided assignments and projects.
- **Instructor**, INFO-I529, Machine Learning Bioinformatics Fall 2024
  Taught methods through hands-on exercises and projects.
- **Assistant Instructor**, DSCI-D351, Big Data Analytics Aug.–Sep. 2024
  Introduced Apache Spark and its data science applications.

Research mentoring

- **Ming Yan, Ian Xia, Evan Gubarev, and Yuntong Pan** Autumn 2026
  CSE 499 Undergraduate Research, University of Washington.
  De novo peptide sequencing with Casanovo and Cascadia.
- **Flynn Keele** Spring 2025
  CSCI-Y 390 Undergraduate Independent Study, Indiana University Bloomington.
  Meta-learning for adaptable pre-trained small-molecule representation models.
  Co-mentored with Haixu Tang.
- **Stephen Kimani** Summer 2024
  IUUR STEM Summer Research Program, Indiana University Bloomington.
  Deep learning for glycan tandem mass spectra prediction.
- **Jordan Spell** Spring 2024
  CSCI-Y 390 Undergraduate Independent Study, Indiana University Bloomington.
  Transfer learning for retention time prediction across experimental conditions.
- **Wenqian Chen** Spring 2024
  M.S. Capstone Research Project, Indiana University Bloomington.
  A workflow for predicting small-molecule retention time across experimental conditions.

Invited talks

1. Beyond database search: machine learning method for small molecule identification. College of Science, San Diego State University, Feb. 11, 2026.
2. Enhanced Structure-Based Prediction of Chiral Stationary Phases for Chromatographic Enantioseparation from 3D Molecular Conformations. Amgen research group (virtual), Oct. 11, 2024.

Conference presentations

1. *De novo* sequencing of chimeric spectra using Casanovo. American Society for Mass Spectrometry (ASMS) annual conference, San Diego, CA, 2026. **Selected oral presentation.**
2. A Task-Specific Transfer Learning Approach to Enhancing Small Molecule Retention Time Prediction with Limited Data. ASMS, Baltimore, MD, 2025. **Selected oral presentation.**
3. Predicting Compositional Fragments of Compounds from Their Tandem Mass Spectra Using Deep Neural Networks. ASMS, Anaheim, CA, 2024. Poster.
4. A Machine Learning Model for Chemical Formula Prediction Using Tandem Mass Spectra of Compounds. ASMS, Houston, TX, 2023. **Selected oral presentation.**
5. Prediction of Molecular Tandem Mass Spectra Using 3-Dimensional Conformers. ASMS, Minneapolis, MN, 2022. Poster.

Professional service

- **Conference reviewing:** KDD (2026, 2027), ACM BCB (2025, 2026), EMBC (2026), NeurIPS AI for Drug Discovery Workshop (2026).
- **Journal reviewing:** ACS Measurement Science Au; Journal of Chemical Information and Modeling; Pharmaceutical Research; Journal of Chromatography A; BMC Genomics; PLOS Computational Biology; PeerJ Computer Science; IEEE/ACM Transactions on Computational Biology and Bioinformatics; BMC Bioinformatics; Chemical Physics Letters; Beilstein Journal of Organic Chemistry.
- **Reviewing under supervision:** RECOMB (2022, 2023, 2025, 2026), ISMB/ECCB (2023, 2025), ACM BCB (2024); Analytical Chemistry; International Journal of Mass Spectrometry (with William Stafford Noble and Haixu Tang).